

Linux — Հրամաններ, Process, Thread

Հարցազրույցի ամբողջական ուղեցույց

Այս ուղեցույցը ընդգրկում է Linux-ի հիմնական հրամանները (ls, cd, pwd, grep, find) և system concepts-ները (process և thread vs process), օրինակներ և interview հարցեր.

1. ls — list

Cucak է talis directory-ի բովանդակությունը.

```
ls                # ներկայիս directory
ls -l            # manramasn (permissions, size, date)
ls -a            # tak (hidden) fayler el (.-ov)
ls -la           # erku-n el
ls -lh           # human-readable size (KB, MB)
ls -lt           # dasavorel est zhamanaki
ls /path/dir     # konkret directory
```

`ls -l` output.

```
-rw-r--r-- 1 user group 1024 Jan 1 file.txt
^type ^perms      ^size   ^date ^name
```

2. cd — change directory

Poxum է ներկայիս directory-ն.

```
cd /home/user      # absolute path
cd folder          # relative (nerkayis-ic)
cd ..              # mek verev (parent)
cd ../..           # erku verev
cd ~               # home directory
cd -               # naxordy (het)
cd                 # home (ařanc argument)
```

3. pwd — print working directory

Cuyc e talis nerkeyis directory-i **liakan (absolute) path-y**.

```
pwd
# /home/user/projects
```

4. grep — search text

Voronum e text (pattern) fayleri mej.

```
grep "hello" file.txt      # "hello" pornox toghery
grep -i "hello" file.txt   # case-insensitive
grep -r "hello" ./         # recursive (bolor subdirectory)
grep -n "hello" file.txt   # tox hamarov
grep -v "hello" file.txt   # HAKAŘaky (ařanc "hello")
grep -c "hello" file.txt   # qanaky (count)
grep -w "cat" file.txt     # amboxj bařy (ne "category")
grep -E "a|b" file.txt     # regex (extended)
```

Pipe-ov (amenataracvac).

```
ps aux | grep chrome      # chrome process-ner
cat log.txt | grep ERROR   # ERROR toghery
```

5. find — search files

Gtnum e fayler/directory-ner criteria-yov.

```
find . -name "*.txt"           # bolor .txt (nerkayis-ic)
find . -type f                 # miayn fayler
find . -type d                 # miayn directory-ner
find . -name "test*"          # "test"-ov sksvox
find . -size +10M              # 10MB-ic mec
find . -mtime -7               # verjin 7 or poxvac
find / -name "config" 2>/dev/null # error-ner tapel
find . -name "*.log" -delete   # gtnel u jnjel
find . -name "*.cpp" -exec grep "main" {} \; # gtnel + kanchel
```

6. Ուրիշ karevor hramanner (bonus)

```
cat file.txt                   # bovandakutyun tpel
less file.txt                  # eja'aj ditel
head -20 file.txt              # a'aji 20 tog
tail -20 file.txt              # verjin 20 tog
tail -f log.txt                # follow (real-time)
cp src dst                     # copy
mv src dst                     # move/rename
rm file                        # jnjel
mkdir dir                      # nor directory
chmod +x script.sh             # permissions
ps aux                         # process-ner
kill PID                       # process spanel
top / htop                     # system monitor
df -h                          # disk space
```

7. Process hũz t

Process-y **ashxatox program** e. Uni ir ařancin memory space, resource-ner, u ařnvazn mek thread.

```
Program (fayl disk-um) --> Process (ashxatox, memory-um)
```

Process-i bnutagreri.

- PID (Process ID) - unikal hamar
- Ařancin memory space (isolated)
- Resource-ner (file handles, sockets)
- State: running, waiting, stopped, zombie
- Parent process (PPID)

```
ps aux          # bolor process-ner
ps -ef          # ayl format
top             # real-time
kill 1234        # PID 1234 spanel
kill -9 1234     # brnov (SIGKILL)
```

8. Thread vs Process

PROCESS:

- Ařancin memory space (isolated)
- Canr (steghcumy tańk)
- Meki myusin chi azdum (crash isolated)
- Communication: IPC (pipe, socket, shared memory)
- Context switch tańk

THREAD:

- KISUM e process-i memory-n (shared)
- Teř (steghcumy ejan)
- Mek thread-i crash -> amboxj process
- Communication: shared memory (unix)
- Context switch ejan

	Process	Thread
Memory	ařancin	kisvac (shared)
Steghcum	taňk (canr)	ejan (teř)
Isolation	ayo	voch
Crash azdecutyun	isolated	amboxj process
Communication	IPC	shared memory
Context switch	taňk	ejan

Analogia

Process = ařancin tun (ařancin xohanoc, senyakner)

Thread = nuyn tan mej mardik (kisum en xohanocy, senyaknery)

9. Interview harcer & patasxanner

ls -l inch e talis?

Manramasn cucak: permissions, owner, size, date, name.

cd .. inch e?

Mek directory verev (parent).

pwd inch e?

Nerkayis directory-i absolute path.

grep -r inch e?

Recursive search bolor subdirectory-nerum.

grep -v inch e?

Hakařak - toghery vor CHEN pornum pattern-y.

find . -name inchpes?

find . -name ".txt" - bolor .txt fayler ner kayis-ic recursive.*

Process inch e?

Ashxatox program, arancin memory space, PID, resource-ner.

Thread vs process?

Process arancin memory (isolated, canr); thread shared memory (mek process, teř). Thread crash -> amboxj process; process crash -> isolated.

ps aux | grep X inch e?

X anunov process-ner gtnel (ps output-y grep-ov filter).

kill -9 inch e?

SIGKILL - process brnov spanel (chi karox anteswel).

Cheat Sheet

```
ls    -> list (ls -la manramasn+hidden, -lh size, -lt time)
cd    -> change dir (cd .. verev, cd ~ home, cd - naxord)
pwd   -> absolute path
grep  -> text search (-r recursive, -i ignore-case, -v invert, -n line, -c count)
find  -> file search (find . -name "*.txt", -type f/d, -size, -exec)
```

PROCESS = ashxatox program, ařancin memory, PID
ps aux | grep X | kill PID | kill -9 (SIGKILL)

THREAD vs PROCESS:

process -> ařancin memory, canr, isolated, IPC
thread -> shared memory, teř, mek process, crash->amboxj

PIPES: ps aux | grep chrome | cat log | grep ERROR